

CI Outcome Prediction Project

Addendum: Graph Signal Reconstruction, Bayesian Imputation,
Post-Op Phenotyping & Subjective Outcomes Analysis

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1 Overview

This addendum documents analyses conducted in December 2024, extending the main report with:

1. **Graph Signal Reconstruction (GSR)** for missing pre-operative speech data
2. **Imputation method comparison:** GSR vs MICE vs Bayesian (Andrea Gabrio approach)
3. **Post-operative phenotype discovery** using spectral clustering
4. **Decline cases analysis** (per Tobias's priority)
5. **Subjective outcomes integration:** SSQ12, THI, IOI-HA
6. **Objective–Subjective disconnect:** 35% discordant patients

1.1 Key Findings

- **GSR limitations:** Pure graph-based imputation fails for edge nodes (FM: $R^2 = -2.4$). MICE outperforms GSR.
- **Bayesian advantage:** Andrea Gabrio-style imputation provides uncertainty (e.g., FM = $42\% \pm 8\%$).
- **CI Equalizer confirmed:** Even with imputed data (N=431), pre-op still doesn't predict post-op ($R^2 \approx 0$).
- **4 post-op phenotypes:** EXCELLENT (36%), GOOD (29%), ATYPICAL (16%), LIMITED (19%).
- **Phenotypes unpredictable:** Pre-op features cannot predict cluster membership (accuracy = 53% = baseline).
- **Decline = High starters:** 22% decline; 69% of top decile ($\text{Mono1}_{\text{pre}} \geq 80\%$) show decline.
- **Decline predictable with Mono1_{pre}:** AUC = 0.79. Without it: AUC = 0.45 (random).
- **Objective $\neq$ Subjective:** Mono1 $\leftrightarrow$ SSQ12 correlation $r = 0.08$ (almost zero!).
- **35% discordant:** Good hearing $\neq$ good quality of life.

2 Graph Signal Reconstruction (GSR) for Missing Data

2.1 Motivation

Pre-operative speech tests have $\sim 43\%$ missingness. We explored GSR as an alternative to standard imputation, leveraging the correlation structure between speech tests.

2.2 Speech Test Graph Structure

Speech tests form a natural chain ordered by difficulty:

$$\text{FM} \longleftrightarrow \text{Num} \longleftrightarrow \text{V08} \longleftrightarrow \text{C12} \longleftrightarrow \text{Mono1}$$

(easiest) (hardest)

Correlation matrix (pre-operative, N=283):

	FM	Num	V08	C12	Mono1
FM	1.00	0.58	0.48	0.48	0.37
Num		1.00	0.60	0.65	0.68
V08			1.00	0.73	0.73
C12				1.00	0.81
Mono1					1.00

Table 1: Pre-operative speech test correlations. Adjacent tests have $r = 0.58$ – 0.81 .

2.3 GSR Algorithm

For each patient with missing values:

1. Identify observed neighbors in the graph
2. Compute weighted average: $\hat{x}_i = \frac{\sum_{j \in N(i)} w_{ij} x_j}{\sum_{j \in N(i)} w_{ij}}$
3. Weights w_{ij} = correlation between tests i and j

2.4 Validation Results (Leave-One-Variable-Out)

Test	N neighbors	R^2	MAE	Assessment
FM_pre	1 (edge node)	-2.40	11.7%	Terrible
Num_pre	2	0.45	10.0%	Weak-moderate
V08_pre	2	0.50	11.1%	Moderate
C12_pre	2	0.66	10.2%	Moderate-strong
Mono1_pre	1 (edge node)	0.46	17.2%	Weak-moderate

Table 2: GSR validation. Edge nodes (FM, Mono1) with only 1 neighbor perform poorly.

Edge node problem: FM and Mono1 are at the ends of the chain with only 1 neighbor each. GSR fails catastrophically for FM ($R^2 = -2.4$, worse than mean imputation).

2.5 Improvement Attempts

Method	FM	Num	V08	C12	Mono1
Chain GSR (baseline)	-2.40	0.45	0.50	0.66	0.46
+ Skip connections	-6.80	-0.41	0.58	0.09	-0.27
+ Correlation weights	-6.35	-0.54	0.51	0.30	-0.21
Fully connected	-9.90	-0.54	0.51	0.30	-0.51
+ Covariates (Age, Audio)	0.14	0.52	0.58	0.70	0.71

Table 3: GSR improvement attempts. Adding weak connections makes performance *worse*. Only adding external covariates helps edge nodes.

Conclusion: Pure GSR only works for middle nodes with multiple strong neighbors. Edge nodes require external information (covariates like age and audiometry).

3 Imputation Method Comparison

3.1 Methods Evaluated

1. **Mean imputation:** Baseline ($R^2 = 0$ by definition)
2. **GSR (chain graph):** Graph-based, neighbors only
3. **MICE:** Multiple Imputation by Chained Equations (IterativeImputer)
4. **Bayesian MVN:** Multivariate normal with PyMC (Andrea Gabrio approach)

3.2 Results

Method	FM	Num	V08	C12	Mono1	Uncertainty?
Mean	0.00	0.00	0.00	0.00	0.00	No
GSR (chain)	-2.40	0.45	0.50	0.66	0.46	No
MICE	0.34	0.56	0.60	0.71	0.73	No*
Bayesian	0.39	0.58	0.61	0.72	0.73	Yes

Table 4: Imputation method comparison (R^2). *MICE can provide uncertainty via multiple datasets.

3.3 Bayesian Imputation: The Andrea Gabrio Approach

Following Andrea Gabrio’s work on missing data in health economics (missingHE package), we implemented Bayesian multivariate normal imputation:

Model:

$$\mathbf{x} \sim \text{MVN}(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \quad (1)$$

$$\boldsymbol{\Sigma} = \text{diag}(\boldsymbol{\sigma}) \cdot \mathbf{R} \cdot \text{diag}(\boldsymbol{\sigma}) \quad (2)$$

$$\mathbf{R} \sim \text{LKJCholesky}(\eta = 2) \quad (3)$$

Key advantage: Posterior distributions for each imputed value.

Variable	Point Estimate	Uncertainty ($\pm$ SD)
FM_pre	42.3%	$\pm 7.8\%$
Num_pre	58.1%	$\pm 12.3\%$
V08_pre	45.7%	$\pm 12.9\%$
C12_pre	61.2%	$\pm 12.1\%$
Mono1_pre	38.4%	$\pm 14.8\%$

Table 5: Bayesian imputation provides uncertainty for each imputed value. Example for a single patient with all speech tests missing.

Why Bayesian matters:

- MICE: FM_pre = 42.3% (point estimate only)
- Bayesian: FM_pre = 42.3% $\pm$ 7.8% (95% CI: [27%, 57%])

Uncertainty propagates to downstream analyses and enables honest reporting in papers.

3.4 Impact on Sample Size

Dataset	Complete Cases	Change
Before imputation	252	–
After MICE	431	+179 (+71%)

Table 6: MICE substantially increases sample size.

3.5 Does Imputed Pre-Op Predict Post-Op Better?

Features	R ² (N=431)
Audio + Age	–0.04
+ Pre-op speech (imputed)	–0.03
Improvement	+0.01

Table 7: Imputation increases N but doesn't improve prediction.

CI Equalizer confirmed: Even with 71% more patients, pre-op features **still don't predict** post-op outcomes. The effect is fundamental, not a sample size issue.

4 Decline Cases Analysis

Per Tobias's priority: "Can we predict which patients will decline?"

4.1 Decline Definition

Patients with $\text{Mono1}_{\text{post}} < \text{Mono1}_{\text{pre}}$ (N=264 with both measurements).

Group	N (%)	$\text{Mono1}_{\text{pre}}$	$\text{Mono1}_{\text{post}}$	Change
Improved	206 (78%)	34%	71%	+37%
Declined	58 (22%)	68%	45%	–23%
Top Decile ($\geq 80\%$)	36	88%	74%	–14%

Table 8: Decline cases have HIGH pre-operative scores (regression to mean).

4.2 Can We Predict Decline?

Features	AUC	Interpretation
Audio + Age only	0.45	Random (no predictive value)
+ $\text{Mono1}_{\text{pre}}$	0.79	Good prediction

Table 9: Decline is predictable **only** with pre-operative WRS.

Answer to Tobias's question: Without $\text{Mono1}_{\text{pre}}$, decline prediction is random ($\text{AUC} = 0.45$). With $\text{Mono1}_{\text{pre}}$, $\text{AUC} = 0.79$. This is because decline = regression to mean.

4.3 Decline Risk by Baseline

Pre-op WRS	N	Decline Risk
Low (0–30%)	89	3%
Mid (30–60%)	73	19%
High (60–100%)	82	52%
Top Decile ($\geq 80\%$)	36	69%

Table 10: Decline risk increases monotonically with baseline. 69% of top-decile patients decline.

5 Post-Operative Phenotype Discovery

5.1 Methods

Spectral clustering on post-operative speech tests (FM, Num, V08, C12, Mono1) with N=378 patients. Gaussian similarity kernel with σ = median pairwise distance.

5.2 Optimal Clusters

k	Silhouette	Calinski-Harabasz	Selected?
2	0.467	321.9	Best by silhouette
3	0.305	264.7	–
4	0.303	224.7	Clinically meaningful

Table 11: k=4 chosen for clinical interpretability.

5.3 Four Post-Operative Phenotypes

Phenotype	N	%	Mono1	FM	Num	Pattern
EXCELLENT	136	36%	84%	98%	99%	Star performers
GOOD	111	29%	64%	92%	96%	Solid outcomes
ATYPICAL	60	16%	42%	60%	85%	FM < Num pattern
LIMITED	71	19%	33%	92%	75%	Need support

Table 12: Four post-operative phenotypes identified by spectral clustering.

5.4 Anomalous Pattern: FM < Num

41% of patients (N=155) show FM < Num, which is unexpected since FM (sentences) should be easier than Num (numbers).

Pattern	N	FM	Num	Diff	Mono1
Normal (FM $\geq$ Num)	223	96.5%	89.3%	+7.2	63.5%
Anomalous (FM < Num)	155	77.9%	94.8%	–17.0	59.5%

Table 13: 41% show reversed FM/Num pattern. Clinical significance to be investigated.

5.5 Can We Predict Phenotype from Pre-Op?

Method	Accuracy	vs. Baseline
Random baseline	52.3%	–
Random Forest (pre-op features)	52.9%	+0.6%

Table 14: Pre-operative features do NOT predict cluster membership.

Key finding: Post-op phenotypes are **real** (distinct outcome patterns) but **unpredictable** before surgery. This confirms the CI Equalizer Effect at the phenotype level.

5.6 Clinical Recommendations by Phenotype

Phenotype	Recommendation
EXCELLENT	Standard follow-up, consider bilateral CI
GOOD	Auditory training may help, FM system for noise
ATYPICAL	Investigate FM<Num pattern, specialized training
LIMITED	Intensive rehab, check mapping, assistive tech, psychological support

Table 15: Phenotype-specific clinical recommendations.

6 Subjective Outcomes Integration

6.1 Available Questionnaires (from PDF: Fragebogen CI-Abklärung)

Questionnaire	Measures	Items	Scale	N (Zurich)
SSQ12	Speech, Spatial, Qualities	12	0–10	59
THI	Tinnitus Handicap	25	Ja/Geleg./Nein	61
IOI-HA	CI/HA Satisfaction	7	varies	201

Table 16: Available subjective outcome questionnaires.

6.2 SSQ12 Structure (Anhang 1)

Item	Question (simplified)	Subscale
1	Understand person with TV on	Speech
2	Follow two speakers at once	Speech
3	Conversation in noisy room	Speech
4	Group of 5 in restaurant	Speech
5	Follow when speaker changes	Speech
6	Locate barking dog	Spatial
7	Judge vehicle distance	Spatial
8	Vehicle approaching/leaving	Spatial
9	Sounds = jumbled mess?	Qualities
10	Identify instruments in music	Qualities
11	Sounds clear (not blurred)?	Qualities
12	Must concentrate hard?	Qualities

Table 17: SSQ12 items. Scale: 0 (cannot do) to 10 (perfect).

6.3 THI Structure (Anhang 2)

25 items measuring tinnitus impact: concentration, understanding, anger, confusion, despair, sleep, social life, frustration, etc.

Scoring: Ja (4) / Gelegentlich (2) / Nein (0). Total: 0–100.

Severity categories:

- Slight: 0–16
- Mild: 17–36
- Moderate: 37–56
- Severe: 57–76
- Catastrophic: 77–100

6.4 Objective–Subjective Correlations

Comparison	Correlation (r)	N	Interpretation
Mono1 ↔ SSQ12	0.08	54	Almost zero!
Mono1 ↔ THI	−0.06	59	Almost zero!
Mono1 ↔ IOI-HA	0.26	190	Weak positive

Table 18: Objective speech scores show minimal correlation with subjective outcomes.

Critical finding: Mono1 ↔ SSQ12 correlation is $r = 0.08$ (almost zero!).

Good objective hearing does NOT equal good quality of life.

6.5 Discordant Patients

	SSQ12 < 5 (poor)	SSQ12 ≥ 5 (good)
Mono1 < 60% (poor)	7	9
Mono1 ≥ 60% (good)	9	27

Table 19: Contingency table. 35% of patients are discordant.

35% of patients are discordant:

- 9 patients: Good Mono1 ($\geq 60\%$), poor SSQ12 (< 5) — hear well, feel bad
- 9 patients: Poor Mono1 ($< 60\%$), good SSQ12 (≥ 5) — hear poorly, feel fine

6.6 THI Severity Distribution

THI Severity	N	%
Slight (0–16)	33	54%
Mild (17–36)	14	23%
Moderate (37–56)	9	15%
Severe (57–76)	4	7%
Catastrophic (77–100)	1	2%

Table 20: THI severity distribution. 23% have moderate or worse tinnitus.

15% have good hearing but high tinnitus: Good Mono1 ($\geq 60\%$) with THI ≥ 36 (moderate+). These patients may explain some of the objective–subjective disconnect.

7 Next Steps & TODO Items

7.1 SQL Queries

TODO:

1. Check if SSQ12 items (1–12) are stored individually
2. Check if THI items (1–25) are stored individually
3. Get data from all 5 CI centers (not just Zurich)
4. Identify CAP/SIR tables for pediatric outcomes

```
-- Check SSQ12 items
SELECT DISTINCT PQS_NAME FROM PAT_QUESTION_SCORE
WHERE PAQ_KEY IN (SELECT PAQ_KEY FROM PATIENT_QUESTIONNAIRE
                  WHERE PAQ_QRE_NAME LIKE '%SSQ%');

-- Check THI items
SELECT DISTINCT PQS_NAME FROM PAT_QUESTION_SCORE
WHERE PAQ_KEY IN (SELECT PAQ_KEY FROM PATIENT_QUESTIONNAIRE
                  WHERE PAQ_QRE_NAME LIKE '%THI%');
```

7.2 If SSQ12/THI Items Available: Spectral Clustering on Subjective Profiles

TODO (if items available):

1. Build 12-node SSQ12 graph (Speech/Spatial/Qualities subscales)
2. Build 25-node THI graph
3. Apply spectral clustering to find **subjective phenotypes**

Why this matters: Two patients with same SSQ12 total = 6 might have very different profiles:

- Patient A: Speech items 8,8,7,8,7 (good), Spatial items 3,2,2 (poor!) — “Good speech, poor spatial”
- Patient B: Speech items 5,4,5,4,5 (moderate), Spatial items 7,7,8 (good) — “Poor speech, good spatial”

Different phenotypes → Different rehabilitation strategies.

7.3 Andrea Gabrio’s Bayesian Approach: Full Implementation

TODO:

1. Use **missingHE** R package (or PyMC implementation)
2. Bayesian imputation with **uncertainty propagation**
3. Test **MAR vs MNAR** assumptions with sensitivity analysis
4. Report **posterior distributions**, not just point estimates

Key references:

- Gabrio et al. (2022). Missing data in trial-based CEA. *Value in Health*.
- missingHE package: <https://github.com/AnGabrio/missingHE>

8 TODO Summary by Source

8.1 Tobias's Priorities (from Notes)

Task	Status	Notes
Decline cases prediction	Done	AUC = 0.79 with Mono1_pre
Decline without Mono1_pre?	Done	AUC = 0.45 (random)
Phenotype discovery (clustering)	Done	4 clusters identified
Composite outcome score	TODO	External vs internal validity?
Manual subgroup phenotyping	TODO	SSD, pre/postlingual, age
Subjective score predictions	TODO	Need SSQ12/THI items
Best-aided vs CI-only testing	TODO	Explain decline cases?
Company-independent models	TODO	Similar to adult/kid question

Table 21: Tobias's priorities from feedback notes.

8.2 Alex Meeting Priorities

Task	Status	Notes
Manual subgroup phenotyping	TODO	Age, SSD, Pre/Postlingual
Categorical prediction (low/avg/good)	TODO	With uncertainty estimates
Patient view	TODO	What is relevant for CI candidates?
Composite score	TODO	Target-independent model
Subjective outcomes	Started	SSQ12/THI integrated, low N
Predict non-users/"bad performers"	TODO	Include decline cases?
Ethics: specify variables	TODO	Ask for specific concerns

Table 22: Alex meeting priorities.

8.3 Technical TODO

Task	Status	Notes
GSR imputation	Done	MICE better, edge node problem
MICE vs Bayesian comparison	Done	Bayesian provides uncertainty
Full Bayesian (missingHE)	TODO	MAR vs MNAR sensitivity
SSQ12/THI item-level analysis	TODO	Awaiting Tuesday SQL
Multi-center data (all 5)	TODO	Increase N for subjective
Spectral clustering on subjective	TODO	If items available

Table 23: Technical TODO items.

9 Summary Tables

9.1 Imputation Recommendation

Use Case	Recommended Method	Reason
Quick exploration	MICE	Fast, good performance
Publication	Bayesian (PyMC / missingHE)	Uncertainty quantification
Sensitivity to MNAR	Andrea's missingHE	Tests missingness assumptions
Graph-based story	GSR (middle nodes only)	Only for Num, V08, C12

Table 24: Imputation method recommendations.

9.2 Key Findings for Paper

Finding	Evidence	Clinical Implication
CI Equalizer Effect	$R^2 \approx 0$	Can't predict post-op from pre-op
4 Post-op phenotypes	Spectral clustering	Real but unpredictable
Decline = High starters	69% of top decile	Counsel high-baseline patients
Objective $\neq$ Subjective	$r = 0.08$	Need dual assessment
35% discordant	Good hearing $\neq$ satisfaction	Different rehab strategies

Table 25: Key findings summary for paper.

10 Questions for Discussion

1. Is the FM < Num pattern (41% of patients) clinically meaningful or artifact?
2. Best-aided vs CI-only testing: Does this explain decline cases?
3. Which composite score criteria: External validity (SSQ12 correlation) or internal validity?
4. Priority: Focus on prediction or phenotyping for initial paper?
5. Should we build separate models for discordant patients?

— End of Addendum —